

Skill3

Question

Apply escape sequences, store command history, navigate previous/next commands, update the input buffer, allocate buffers dynamically and manage memory safely.

Simple C Program

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>

int main() {
    char **history = malloc(5 * sizeof(char *));
    char command[100];
    int count = 0;

    while (count < 5) {
        printf("myshell> ");
        fgets(command, sizeof(command), stdin);
        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
            break;

        history[count] = malloc(strlen(command) + 1);
        strcpy(history[count], command);
        count++;
    }

    printf("\nCommand History:\n");
    for (int i = 0; i < count; i++) {
        printf("%d: %s\n", i + 1, history[i]);
        free(history[i]);
    }

    free(history);
    return 0;
}
```

OUTPUT:

```
pothubhavana@DESKTOP-16SJJUA:~$ nano skill3.c
pothubhavana@DESKTOP-16SJJUA:~$ gcc skill3.c
pothubhavana@DESKTOP-16SJJUA:~$ ./a.out
myshell> bhavana
myshell> ls
myshell> pwd
myshell> date
myshell> hello

Command History:
1: bhavana
2: ls
3: pwd
4: date
5: hello
pothubhavana@DESKTOP-16SJJUA:~$ |
```